EMMANUEL BOAKYE

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GitHub

RESEARCH INTERESTS

- Recommender Systems and Learning to Rank
- Retrieval and Ranking for Large-Scale, Sparse Data
- Large Language Models and Foundation Models for Recommendation
- Machine Learning for Health and Clinical Data
- Explainability and Trustworthy AI

EDUCATION

MSc Data Science **Sep 2024 – Sep 2025**
Loughborough University *Loughborough, UK*

Relevant Modules: Advanced Machine Learning, Statistical Analysis, Practical AI, Data Analysis & Visualization, Programming for Data Scientists, Data Mining, Research Methods.

BA Social Science - Economics (Upper Second Class Honours) **Aug 2019 – Aug 2023**
University of Cape Coast (UCC) *Cape Coast, Ghana*

Relevant Coursework: Econometrics, Advanced Microeconomics & Macroeconomics, Data Analysis, Mathematics for Economists I & II, Statistics I & II

RESEARCH EXPERIENCE

Dissertation Follow-Up Research **April 2026 – July 2026**
Title: *Retrieval or Ranking? Diagnosing the Dominant Bottleneck in a Hybrid Two-Staged Recommender on a Sparse Product Catalogue.* (Link) (Paper)

- Built a two-stage recommender (SVD, SBERT, item-to-item to LightGBM LambdaRank) on Amazon Electronics 2023 that separates candidate retrieval from ranking (79,838 users, 48,154 items), with out-of-fold candidate generation so no evaluation information could reach training.
- Reported that a simple popularity baseline (HR@10=0.61) outperformed the full hybrid system (HR@10=0.30) and treated that as a finding.
- Showed that training the ranker on retrieved candidates rather than observed pairs raised isolated HR@10 from 0.17 to 0.52 and that 84% of its decisions were from item popularity.

MSc Dissertation **May 2025 – Sep 2025**
Title: *Item Recommender System Using Machine Learning* **Supervisor:** Dr. Shaheen Syeda Fatima
Loughborough University *Loughborough, UK*

- Built a hybrid recommender on 200,000 Amazon electronics reviews, stacking SVD, TF-IDF similarity, and a review-text CNN under an XGBoost meta-learner to predict ratings.
- Evaluated with regression metrics (RMSE, MAE, R^2) and ranking metrics (Precision, Recall, NDCG@K).
- Its headline results (R^2=0.89) proved to be inflated by data leakage which motivated the follow-up study.

Postgraduate Research Intern, HDRUK **Jan 2025 – April 2025**
Title: *Predicting Survival in Critically ill Patients using Classical and Machine Learning Models* (Link)
Loughborough University *Loughborough, UK*

- Benchmarked a Cox proportional hazards model against Random Survival Forest and Gradient Boosted Survival for time-to-death prediction in critically ill ICU patients.
- Built an end-to-end modeling pipeline to handle right-censoring and missing clinical values, excluding outcome-derived columns to avoid leakage.
- Evaluated with the concordance index, time-dependent AUC, and integrated Brier score rather than a single accuracy figure and presented the work as a research poster.

PUBLICATIONS & WORKING PAPERS

Boakye, E. (2026) *Retrieval or Ranking? Diagnosing the Dominant Bottleneck in a Hybrid Two-Staged Recommender on a Sparse Product Catalogue.* **(Preprint on Zenodo, not peer-reviewed).** (Paper)

SELECTED PROJECTS

Vellum-NLQ — Governed Natural-Language Query Layer for Insurance Claims (Link) **May 2026**

- Built a text-to-SQL layer that lets a non-technical user query medical insurance claims data in plain English.
- Added a governance boundary so generated queries run only against permitted tables and columns, which keeps the system safe on regulated data.
- Returned the generated SQL alongside the answer, so a reviewer can audit what the model actually ran.

EDGAR RiskLens — Explainable Risk and Sentiment Analysis of SEC Filings (Link) **Mar 2026 – Apr 2026**

- Built an NLP system that reads public SEC filings and scores them for sentiment, risk, and uncertainty using finance-specific word lists, with every score traceable to matched terms and cited excerpts.
- Broke overall risk into categories such as litigation, regulatory, liquidity, market, cybersecurity, and operational, and tracked risk across a company's recent filings to show whether it is rising or falling.
- Parsed filing sections, supported both US issuer forms (8-K, 10-Q, 10-K) and foreign issuer forms (6-K, 20-F).

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

Machine Learning: Supervised & Unsupervised Learning, gradient boosting (XGBoost, LightGBM), deep learning (CNN, Transformers), anomaly detection, feature engineering

NLP & LLMs: Hugging Face Transformers, Sentence-Transformers, LangChain, RAG, vector databases

Statistics & Methods: Regression, classification, Clustering, Bayesian Methods, Time Series, Model Evaluation

Frameworks & Tools: PyTorch, scikit-learn, Pandas, NumPy, PostgreSQL, Git

TEACHING & MENTORING

Track Lead (Mentor Me Collective) **April 2026 – Jul 2026**
Grow with Google *Remote, USA*

- Led a group of 12 participants through the Google Advanced Data Analytics Certification program.
- Organized weekly sessions, facilitated discussions on statistical concepts and machine learning.
- Supported members with Python (Pandas, NumPy, scikit-learn) and data storytelling

PROFESSIONAL EXPERIENCE

Data Scientist Intern **Nov 2025 - Feb 2026**
Fildata *Manchester, UK*

- Built a fraud detection model on transaction data using gradient-boosted trees and anomaly detection, tuned to flag suspicious activity while keeping false positives manageable.
- Developed a customer churn prediction model to identify at-risk clients and support retention decisions.

National Service Personnel (Data Quality) **Oct 2023 – Sep 2024**
St. Louis College of Education *Ghana*

- Built statistical checks to detect anomalies and inconsistencies across student records of more than 18,000 entries, improving the reliability of institutional reporting.
- Developed dashboards and summary reports that gave staff clearer visibility into enrollment and student data for decision-making.

AWARDS & HONOURS

- Talent Match Recognition, Loughborough University (2025)
- AI in Education Hackathon, DigiTech @ Loughborough University 2025 (EDI inclusivity challenge using AI)